

Module 6: Exponential Functions – Prequiz

Name:

Date:

Four short questions. Interpret the initial value and repeated multiplicative change.

1. For $A(t) = 400(1.06)^t$, state the initial value and the percent growth rate.
2. A population is modeled by $P(t) = 250 \cdot 2^{t/5}$. Find $P(10)$.
3. A substance begins with 960 grams and has a half-life of 4 hours. How much remains after 12 hours?
4. A continuously growing quantity is modeled by $B(t) = 500e^{0.03t}$. Find $B(5)$ to three decimal places.

Module 6: Exponential Functions – Weekly Check-In

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Three questions. Show enough work to make the repeated multiplicative structure visible.

1. Reading an exponential model.

A population is modeled by

$$P(t) = 450(1.08)^t,$$

where t is measured in years.

(a) State the initial population and the percent growth rate.

(b) Find $P(6)$ to three decimal places.

2. Half-life.

A medication begins with 640 mg in the body and has a half-life of 6 hours.

(a) Write an exponential model for the amount $A(t)$ after t hours.

(b) Find $A(9)$ to three decimal places and $A(18)$ exactly.

Module 6: Exponential Functions – Weekly Check-In, continued

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3. Continuous exponential growth.

A culture begins with 300 cells and is modeled by

$$C(t) = 300e^{0.08t},$$

where t is measured in hours.

- (a) Explain what the numbers 300 and 0.08 tell you about the model.

- (b) Find $C(5)$ to three decimal places.

- (c) Is the quantity growing or decaying? Explain how the formula tells you.